

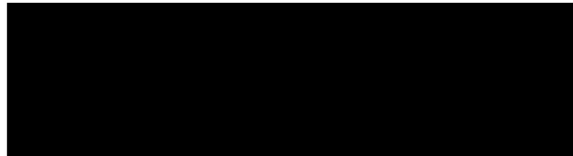
Chapter 8

MORE REPETITION

Before we begin

What is the **assignment operator**?

Answer: The equal sign



Question 2

What is an **assignment statement**?

Answer 2

A statement used to assign a value to a variable while a program is running.

Example:

```
age = age + 10;
```

Question 3

When can you **not** use an assignment statement?

Answer 3a

You **cannot** use an assignment statement to assign a value to a **named constant** because the contents of a named constant cannot be changed during runtime.

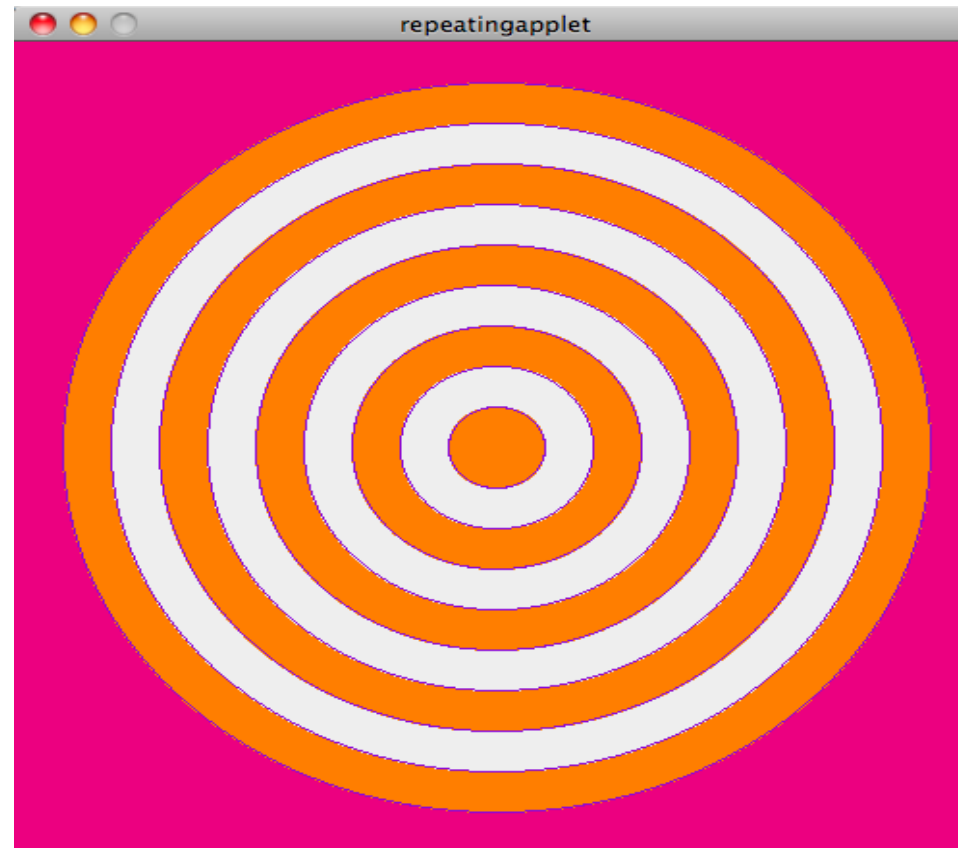
Answer 3b

You **cannot** use an assignment statement when you should be using an comparison. For example,

While (x == 1)

This is a comparison, not an assignment.

Chapter 8 – more on the **repetition** structure



Question 4

If you enter 5, 8, 9 and -1, how many times will the **condition** in the algorithm below be evaluated?

1. enter number
2. repeat while (**number is greater than or equal to 0**)
 - Display number
 - Enter number
 - End repeat

Answer 4

4

Question 5

If you enter -4, how many times will the **condition** in the algorithm below be evaluated?

1. enter number

2. repeat while (**number is greater than or equal to 0**)

- Display number
- Enter number
- End repeat

Answer 5

1

Question 6

If you enter 5, 8, 9 and -1, how many times will the **condition** in the algorithm below be evaluated?

1. enter number

2. repeat

- Display number
- Enter number
- End repeat while (**number is greater than or equal to 0**)

Answer 6

3

Question 7

If you enter 0 and -4, how many times will the **condition** in the algorithm below be evaluated?

1. enter number

2. repeat

- Display number
- Enter number
- End repeat while (**number is greater than or equal to 0**)

Answer 7

1

What was the difference in the condition evaluation between the previous loops?

enter number

repeat while (**number is greater than or equal to 0**)

- Display number
- Enter number
- End repeat

enter number

repeat

- Display number
- Enter number
- End repeat while (**number is greater than or equal to 0**)

Answer: What was the **difference** in the condition evaluation between the previous loops?

enter number

repeat while (**number is greater than or equal to 0**)

- Display number
- Enter number
- End repeat

Pretest

enter number

repeat

- Display number
- Enter number
- End repeat while (**number is greater than or equal to 0**)

Posttest

Also called:

Top driven

Bottom driven

So, now we have-----

Pretest loops (top driven)

Posttest loops (bottom driven)

While statements

For statements

And now a new statement -----

do while statement

This is for a posttest loop only!

```
do //begin loop
{
}
while (condition);
```

Question 9

Write a C++ *while* statement that process the body of a posttest loop as long as the value in the ***inStock*** variable is greater than the value in the ***reorder*** variable.

Answer 9

do

{

}

while (inStock > reorder);

Question 10

Write a C++ ***while*** clause that processes the body of a posttest loop as long as the value in a char variable named ***letter*** is either Y or y. Use the toupper function.

Answer 10

do

{

}

while (toupper(letter) == 'Y');

Nested Repetition Structures

You can place a loop within a loop.

You will have an outer loop and an inner loop.

You can mix pretest and posttest loops!

Who decides what to use?

The programmer decides after studying the problem specification.

How does this work?

```
for (int line = 1; line < 4; line += 1)
{
    cout << "*" << endl;
}
//endfor
```

Answer for Previous How does this work?

*

*

*

Now a twist! What is this doing?

```
for (int line = 1; line < 5; line = line + 1);
```

```
{  
    for (int numAsterisks = 1; numAsterisks < 4; numAsterisks += 1)  
    {  
        cout << "*";  
    }  
    //end for  
    cout << endl;  
} // end for
```

Answer for Previous Slide

Question 11

A nested loop can be a pretest loop only?

True or false?

Answer 11

false

Question 12

Consider a clock's hour and minute hands only. The hour hand is controlled by a(n) _____ loop, while the minute hand is controlled by a(n) _____ loop.

- a. outer, nested b. nested, outer

Answer 12

Consider a clock's hour and minute hands only. The hour hand is controlled by a(n) outer loop, while the minute hand is controlled by a(n) nested loop.

New function for you!

pow function

For exponentiation

Raises a number to a power and returns the result as a double number.

Looks like this!

`pow(x, y)`

`double cube = 0.0;`

`cube = pow(4.0, 3);`

Therefore, `cube = 64`

New function, new include

`#include <cmath>`

Now you know 3 includes:

```
#include <iostream>
```

```
#include <iomanip>
```

```
#include <cmath>
```


My advice?

Put all the *includes* in your programs whether you need them or not.

Question of the day?

What are *fundamental data types*?

Answer of the day

Fundamental data types are the basic data types built into the C++ language.

Sometimes called primitive data types.

Examples: int, double, float, char, bool, short

String is not one!